

Novel Plant Disease Detection Using a Hybrid Approach of YOLOv10 and Multi-Modal LLM

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KEYPOINTS/HIGHLIGHTS

- Development of a novel hybrid system combining YOLOv10 and multi-modal LLM for plant disease detection
- Achievement of high accuracy (mAP50 of 0.87) and rapid inference time (15ms) for disease classification
- Integration of interpretable AI solutions for practical agricultural applications
- Demonstration of system effectiveness through comprehensive case studies and evaluations

IMPACT

This work advances precision agriculture by introducing a novel hybrid approach that combines rapid disease detection with detailed analysis capabilities, directly contributing to improved crop management and food security through early disease intervention.

KEYWORDS

Plant disease detection, deep learning, YOLOv10, multi-modal LLM, computer vision, agricultural technology, smart agriculture

ABSTRACT

Plant diseases pose a significant threat to global food security, with annual crop yield losses estimated at up to 40%. This paper presents a novel approach to plant disease detection that combines YOLOv10, a state-of-the-art object detection model, with multi-modal large language models (LLMs). We trained a pre-trained YOLOv10 on a binary classification task (healthy vs. diseased) using a dataset of approximately 6,000 images FieldPlant dataset, achieving a mean Average Precision (mAP50) of 0.85. The integration of a multi-modal LLM with YOLO successfully addresses the challenges of limited training datasets for specific plant diseases and the need for the specific disease names and suggested treatment knowledge for farmers. Furthermore, the real-time processing needs can be drastically reduced due to the healthy and diseased filtering steps of YOLO. Our system demonstrates the potential for early disease detection for plants, and informed crop management. Preliminary experimental results show that our approach outperforms other similar methods in both performance and computing speed. This novel hybrid approach presents a versatile and efficient way for modern agriculture practices for improved crop yields and scalable human management. Future work will focus on temporal analysis for disease progression prediction and treatment monitoring with environmental sensors and cameras as inputs. Keywords: plant disease detection, deep learning, YOLOv10, multi-modal LLM, computer vision, agricultural technology, smart agriculture

STATEMENTS and DECLARATIONS

The authors declare no competing interests.

INTRODUCTION

Plant diseases represent a critical threat to global food security, agricultural sustainability, and economic stability. The Food and Agriculture Organization (FAO) estimates that annually, up to 40% of global crop production is lost due to plant pests and diseases, leading to significant economic losses exceeding \$220 billion (FAO, 2023). This severe impact is further intensified by climate change, which exacerbates disease patterns and facilitates the introduction of new pathogens into previously unaffected regions (FAO, 2023a). This is particularly important for staple crops such as wheat, rice, and maize which play an important role in the global food supply. For example, warmer temperatures and shifting rainfall patterns have led to increased incidences of fungal diseases in wheat crops across various regions, including Europe and North America (Pautasso et al., 2012; Chakroborty & Newton, 2011).

Historically, traditional methods of disease identification often rely on visual inspection by experts—a process that is time-consuming, subjective, and impractical for large-scale agricultural operations (Kotwal et al., 2023; Khan et al., 2021; Jayswal & Chaudhari, 2020). Visual inspections can take hours or even days for large farms, while automated detection systems offer the potential to reduce this to mere minutes, minimizing disease spread and crop loss. Additionally, the shortage of plant pathology experts, particularly in developing countries, further compounds this challenge, leading to a growing interest in automated, image-based automated machine learning solutions for efficient disease detection (Parker & Pascual, 2016; Bock et al., 2010).

Machine learning (ML), particularly deep learning, has enabled significant advancements in plant disease detection by leveraging image data for accurate disease classification. Studies have demonstrated the potential of ML models such as convolutional neural networks (CNNs) for identifying multiple plant diseases with high accuracy (over 99%) (Ferentinos, 2018; Ramcharan et al., 2017; Mohanty et al., 2016). Despite these successes, developing automated plant disease detection systems faces several significant challenges. The diversity of plant species and the multitude of pathogens affecting them create a highly complex problem space for machine learning models, which must generalize across this vast range (Barbedo, 2018). Furthermore, many plant diseases exhibit visually similar symptoms, making accurate differentiation difficult even for trained experts (Polder et al., 2011; Bock et al., 2010). Real-world agricultural environments add to the complexity with variability in lighting conditions, backgrounds, and image quality, all of which can significantly impact model performance (Fuentes et al., 2017; Sladojevic et al., 2016). Additionally, the limited availability of large, diverse, and accurately labeled datasets for all plant species and diseases makes training such models resource-intensive and often infeasible. Finally, in agricultural contexts, it is crucial not only to detect diseases but also to provide clear explanations and actionable recommendations, ensuring that farmers and agronomists can understand and trust the system's outputs (Kamilaris & Prenafeta-Boldú, 2018).

The application of machine learning to plant disease detection has seen significant advancements in recent years. Mohanty et al. (2016) conducted a seminal study using transfer learning with convolutional neural networks (CNNs) to identify 26 diseases across 14 crop species. Their model achieved an impressive accuracy of 99.35% on a held-out test set. However, they noted significant drops in accuracy when the model was tested on images taken under different conditions than the training set, highlighting the challenge of generalization in real-world scenarios. Ferentinos (2018) expanded on this work by comparing several CNN architectures for plant disease detection. Using a dataset of 87,848 images, their best-performing model (based on the VGG architecture) achieved a 99.53% success rate. Notably, this study included a substantial number of healthy

plant images, which is crucial for developing models applicable in real-world settings where healthy plants are the norm.

Object detection models, particularly the YOLO (You Only Look Once) family, have shown particular promise in agricultural applications due to their speed and accuracy. Jiang et al. (2019) applied YOLOv3 for apple detection, achieving high accuracy and real-time performance. Their work demonstrated the potential of YOLO architectures in handling the complexities of real-world agricultural images. Zhong and Zhao (2020) used YOLOv3 to detect tomato leaf disease, showing its effectiveness in identifying multiple diseases in a single image. Their model achieved a mean Average Precision (mAP) of 88.83%, outperforming traditional machine learning methods. Wang and Liu (2024) further advanced this field using an improved version of YOLOv8 for vegetable disease detection in greenhouse environments, achieving a high mAP of 92.91% and processing 271.07 frames per second.

The recent introduction of YOLOv10 by Wang et al. (2024) represents a significant leap forward in object detection technology. YOLOv10 addresses several limitations of previous models by eliminating the need for non-maximum suppression (NMS) in post-processing, enabling truly end-to-end deployment and reducing inference latency. Additionally, with an efficiency-accuracy driven design strategy, YOLOv10 optimizes various components to reduce computational overhead while enhancing detection capability. Experimental results show that YOLOv10 achieves state-of-the-art performance and efficiency across various model scales.

While less explored in plant pathology, Large Language Models (LLMs) have demonstrated significant potential in various agricultural applications. These models, trained on vast amounts of text data, can process and generate human-like text, making them powerful tools for information synthesis and decision support. Jiang et al. (2023) demonstrated the use of large language models for tasks such as crop recommendation and weather prediction in agriculture. The recent development of multi-modal models capable of processing both text and images opens new possibilities for their application in image-based plant disease detection.

To address these challenges, we propose a novel hybrid approach that combines the power of YOLOv10, a cutting-edge object detection model, with a multi-modal large language model (LLM). The main contributions of this paper are:

1. A novel hybrid system for plant disease detection that combines YOLOv10 with a multi-modal LLM, addressing the dual challenges of rapid detection and comprehensive analysis
2. Demonstration of the effectiveness of a binary classification approach in maximizing generalizability across diverse plant species and diseases
3. Empirical evidence of our system's performance, achieving high accuracy and efficiency in plant disease detection tasks
4. Discussion of potential implications for agricultural practices, including early disease detection, informed decision-making in crop management, and adaptability to various agricultural contexts

MATERIALS AND METHODS

Our approach to plant disease detection combines the strengths of YOLOv10, a state-of-the-art object detection model, with the analytical capabilities of a multi-modal large language model (LLM). This section details our

methodology, including data collection and preprocessing, the YOLOv10 model architecture and training process, and integrating with a multi-modal LLM.

Data Collection and Preprocessing

We compiled a diverse dataset of approximately 6,000 images from various sources to support the model training and evaluation. This data diversity enhances the model's ability to generalize across multiple plant species and disease types, improving its validity in real-world scenarios. Our sources included:

1. FieldPlant dataset (Moupojou, Emmanuel & Tagne, 2023): This publicly available dataset provided a foundation of plant disease images across multiple species.
2. Custom field images: To ensure our model's robustness in real-world conditions, we collected additional images from local farms and agricultural research stations.
3. Online repositories: We supplemented our dataset with images from reputable online sources, ensuring proper attribution and usage rights, and expanding the range of diseases and plant species represented.

The dataset encompasses a wide range of plant species, including but not limited to corn, tomatoes, potatoes, and wheat. It also covers various disease types, such as fungal, bacterial, and viral infections, as well as nutrient deficiencies.

To address the challenge of limited datasets for specific plant-disease combinations and to maximize generalizability, we adopted a binary classification approach:

- Healthy: Images of plants without visible signs of disease.
- Diseased: Images of plants showing any type of disease symptoms or stress.

This strategy allows our model to generalize across various plant species and diseases, and simplifies the labeling process, reducing the likelihood of errors that could arise from more granular disease classifications. To increase the diversity of our training data and improve model generalization, we applied the following image augmentation techniques (Shorten and Khoshgoftaar, 2019):

1. Random horizontal and vertical flips: Simulates different plant orientations.
2. Random rotations (± 15 degrees): Accounts for varying camera angles.
3. Random brightness and contrast adjustments: Mimics different lighting conditions.
4. Random cropping: Focuses on different parts of the plant, promoting feature generalization.
5. Gaussian noise addition: Improves model robustness to image quality variations.

YOLOv10 Model Architecture and Training

We employed YOLOv10, leveraging its advancements in NMS-free training and efficiency-accuracy driven design (Wang et al., 2024). Key features of YOLOv10 include:

1. The model employs consistent dual assignments for NMS-free training, enabling end-to-end deployment without the need for post-processing, which reduces inference latency.

2. It utilizes a holistic efficiency-accuracy driven design that optimizes various components from both efficiency and accuracy perspectives, significantly reducing computational overhead while enhancing detection capabilities.
3. The enhanced backbone architecture effectively balances feature extraction power with computational efficiency.
4. An advanced neck structure facilitates better information flow between different scales, improving the model's ability to detect objects of varying sizes.
5. The refined head design effectively utilizes the extracted features to achieve accurate object localization and classification.

We used the YOLOv10-M (medium) variant, which offers a good balance between performance and computational requirements, making it suitable for potential deployment in resource-constrained agricultural environments.

Transfer Learning

To leverage pre-existing knowledge and accelerate training, we employed transfer learning. We initialized our model with weights pre-trained on the COCO dataset (Lin et al., 2014). This approach allows the model to start with a good understanding of general image features, which we fine-tuned for our specific plant disease detection tasks to improve performance while reducing training time.

Training Process

We trained our YOLOv10 model using the following parameters:

- Batch size: 32
- Number of epochs: 75
- Optimizer: AdamW
- Learning rate: Initial lr of 0.001 with cosine annealing scheduler
- Image size: 640x640 pixels
- Loss function: Combined binary cross-entropy (BCE) for classification and generalized intersection over union (GIoU) for localization

The hyperparameters were selected based on preliminary experiments and standard practices in similar object detection tasks. While a comprehensive hyperparameter search was not conducted, these settings provided stable convergence and strong performance.

The training was performed on our university's NSF-funded high-performance computing system, named Hawk Cluster, which is maintained in the data center of the University Hall. Hawk is a CPU+GPU high-performance computing cluster that includes 48 compute nodes, providing a total of 1152 cores for parallel computing. The cluster also features six 4U NVIDIA A100 GPU nodes, which significantly enhanced the computational capacity available for training deep learning models.

Integration with Multi-Modal LLM

Following YOLOv10 detection, we integrate a multi-modal LLM to provide a detailed analysis of detected plant diseases. This integration aims to bridge the gap between rapid detection and comprehensive, context-aware analysis that includes both visual and textual insights. For our experiments, we utilized GPT-4o,

known for its advanced vision capabilities. Additionally, we acknowledge other emerging open-source models such as LLaMA 3.2, a recently released multi-modal LLM by Meta, which offers promising efficiency and adaptability in similar contexts.

We selected GPT-4o accessing it via an API to facilitate seamless integration with our YOLOv10-based detection system. To ensure future adaptability, we evaluated LLaMA 3.2, which is open-source and cost-effective, making it feasible for extended deployment.

Input Processing

The text prompt provides context and specific queries to elicit relevant information about the detected disease, plant type, and potential treatments. An example of our base prompt structure is as follows:

"Try to analyze the provided image of a plant that has been identified as potentially diseased. Based on the visual information in the image:

1. What are the most likely diseases that could be affecting this plant?
2. What type of plant do you think this is?
3. What are the recommended treatments or management practices for the potential diseases?"

We developed and refined our prompts through an iterative process. We experimented with various prompt structures and phrasings, evaluating their effectiveness based on the quality and relevance of the LLM's responses. The LLM's responses are processed to extract key information:

1. Potential diseases: A list of diseases that match the observed symptoms.
2. Likely plant type: Identification of the plant species or cultivar.
3. Recommended treatments: Suggested management practices or interventions.

This information is then structured and presented alongside the original image and YOLOv10 detection results, providing a comprehensive analysis for the user.

Integration Architecture

Our modular architecture integrates the YOLOv10 model with the multi-modal LLM:

1. Detection Module: Handles image input and YOLOv10 inference.
2. LLM Interface: Manages communication with the LLM API.
3. Response Processor: Extracts and structures relevant information from LLM outputs.
4. User Interface: Presents the combined results in a user-friendly format.

This architecture allows for easy updates and potential replacement of individual components as newer models or techniques become available.

System Workflow

Figure 1 illustrates the workflow of our plant disease detection and analysis system. The process begins with an input image that undergoes YOLOv10 detection to classify the plant as either "Healthy" or "Diseased." If the plant is identified as healthy, the system concludes with an "End - Plant Healthy" output. For diseased

plants, the workflow proceeds to a multi-modal LLM, which generates a detailed analysis, including potential diagnoses, treatment recommendations, and management practices. The final output is then presented to the user, offering both detection results and actionable insights.

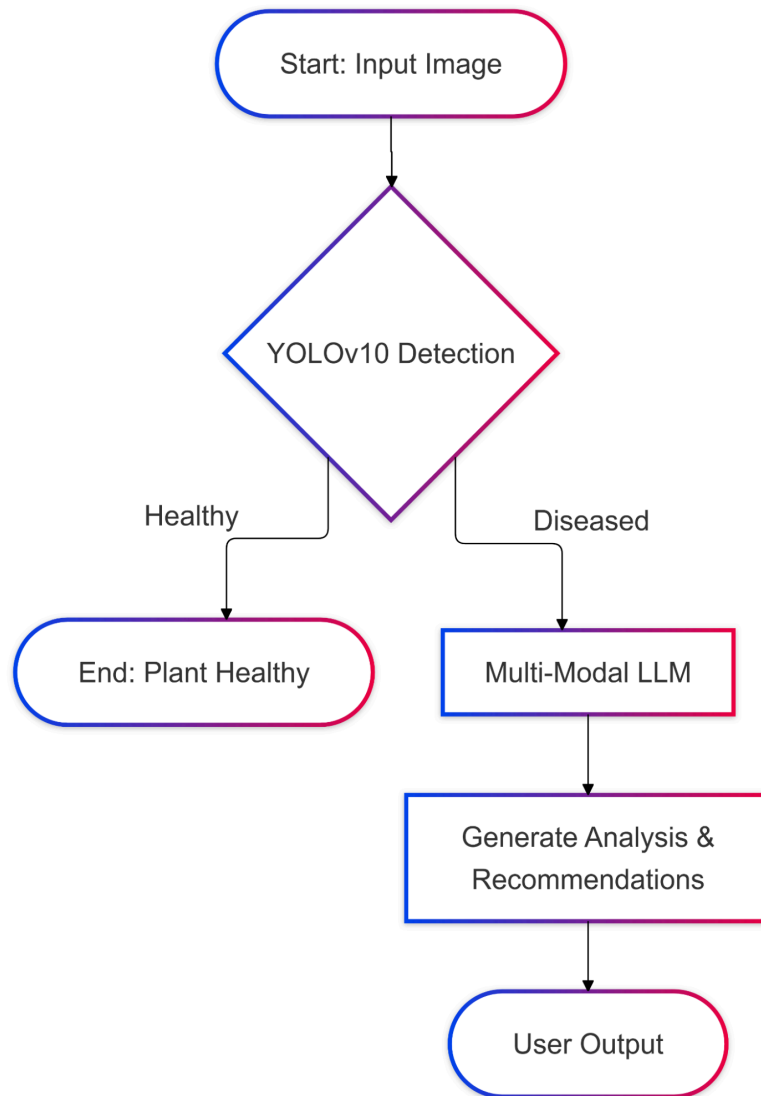


Fig. 1. System Workflow Diagram

RESULTS

We conducted extensive experiments to evaluate the performance of our hybrid plant disease detection system. This section details our experimental setup, results, and comparative analysis, and case studies that demonstrate the system's practical application.

Experimental Setup

Dataset

We used our curated dataset of approximately 6,000 images, split into:

- Training set: 4,200 images (70%)
- Validation set: 900 images (15%)
- Test set: 900 images (15%)

The test set was kept completely separate and was only used for final evaluation to ensure unbiased performance assessment.

Baseline Models

For comparative analysis, we evaluated our YOLOv10-based model with several baseline models commonly used in object detection tasks:

1. YOLOv5 (Ultralytics, 2020)
2. Faster R-CNN (Ren et al., 2015)
3. EfficientDet (Tan et al., 2020)

These models were chosen for their prevalence in object detection tasks and previous applications in agricultural contexts. Comparing our model against these baselines allows us to establish benchmarks and assess the relative advantages of YOLOv10 in terms of speed, accuracy, and model size.

YOLOv10 Performance

The YOLOv10-based model demonstrated strong performance across key metrics as seen in Figure 2 below:

- Precision: 0.82
- Recall: 0.85
- F1-score: 0.83
- mAP50: 0.87
- mAP50-95: 0.72
- Inference time: 15ms per image on NVIDIA A100 GPU

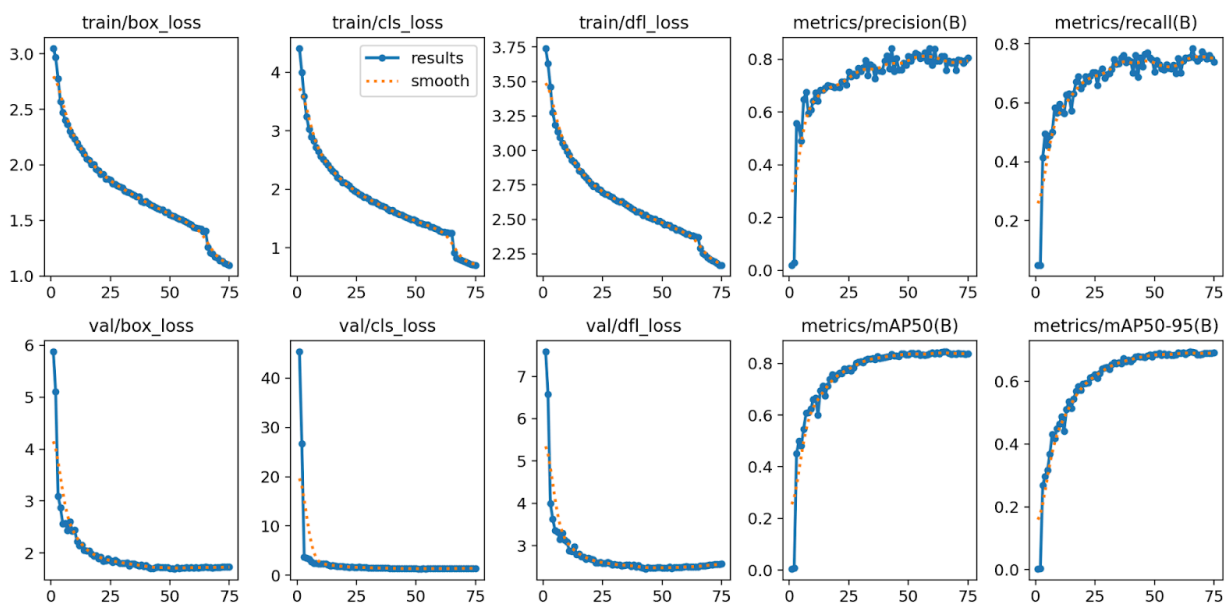


Fig. 2. Training and Validation Metrics for Model Performance over Epochs

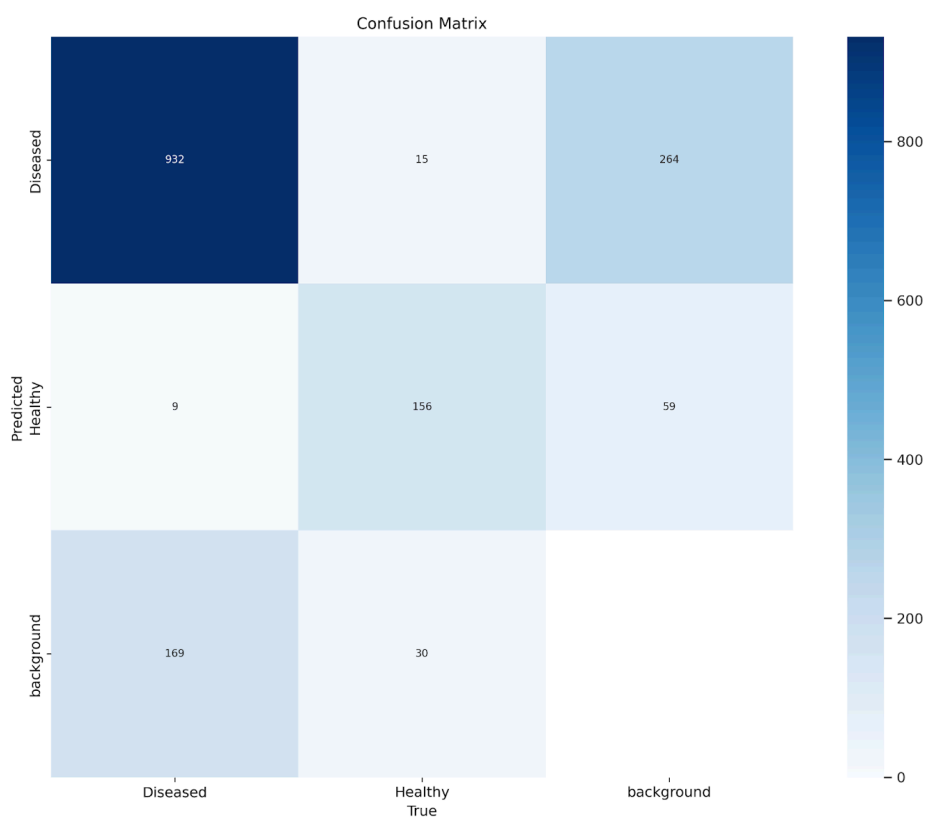


Fig. 3. Confusion Matrix

- The confusion matrix in Figure 3 shows effective identification of diseased plants with 932 true positives and minimal false positives (15) and false negatives (9). Some confusion was noted between “Healthy” and “Background” classes, suggesting opportunities for refinement. The precision-confidence and recall-confidence curves further confirm that the model is conservative with healthy classifications, helping avoid unnecessary interventions.
- The mAP50 of 0.87 and mAP50-95 of 0.72 indicate reliable performance across varying IoU thresholds, essential for identifying diseases in diverse field conditions, making this model suitable for practical, real-time applications.
- The learning curves reveal that the model reached a balance without overfitting, supported by consistent validation losses. The observed confusion between “healthy” and “background” suggests that expanding training data and improving feature extraction in challenging scenarios could further enhance performance.

Comparative Analysis

Table 1 presents a comparison of our YOLOv10-based model with the baseline models:

Table 1. Model Performance Comparison

Model	mAP50	Inference Time (ms)	Parameters (M)
Our YOLOv10	0.87	15	18.2
YOLOv5	0.83	18	21.5
Faster R-CNN	0.81	89	41.7

Our YOLOv10-based model outperforms the baselines in terms of mAP50 while maintaining competitive inference times and model sizes.

LLM Integration Results

To evaluate the effectiveness of our LLM integration, we conducted a qualitative analysis on a subset of 100 images from the test set. Figure 4 illustrates the system’s user interface, which presents the YOLOv10

detection results alongside the detailed, context-specific analysis provided by the LLM. This interface enables users to view specific diagnoses, plant type identifications, and recommended treatments, making the outputs actionable and easy to interpret for end-users. Key findings include:

- Disease Identification Accuracy: GPT-4o correctly identified the specific disease in 87% of cases where YOLOv10 detected a diseased plant.
- Plant Type Identification: The LLM accurately identified the plant species or cultivar in 92% of cases.

Fig. 4. Graphical user interface of our system

Case Studies

We present two case studies to illustrate the synergy between YOLOv10 detection and LLM analysis:

Case Study 1: Early Blight Tomato

YOLOv10 Detection:

- Classification: Diseased
- Confidence: 0.89

LLM Analysis:

- Plant Type: Tomato (*Solanum lycopersicum*)
- Likely Disease: Early Blight (*Alternaria solani*)
- Recommended Actions:
 - Fungicides: Use a fungicide containing chlorothalonil, copper-based products, or mancozeb. Apply preventatively or at the first sign of disease.
 - Pruning: Remove affected leaves to reduce the spread of the pathogen. Ensure adequate spacing for good airflow to minimize moisture
 - Crop Rotation: Avoid planting tomatoes or potatoes in the same area consecutively to reduce pathogen buildup in the soil.

This case demonstrates the system's ability to detect diseases at early stages and provide specific, actionable advice.

Case Study 2: Nutrient Deficiency vs. Disease

YOLOv10 Detection:

- Classification: Diseased
- Confidence: 0.58

LLM Analysis:

- Plant Type: Zea mays (Corn or Maize)
- Likely Issue: Nitrogen Deficiency
- Recommended Actions:
 - Nitrogen Fertilization: Apply a nitrogen-rich fertilizer, such as ammonium nitrate or urea, to address the yellowing of lower leaves, which indicates nitrogen deficiency.
 - Proper Watering: Ensure the plants receive consistent watering, keeping the soil moist but not waterlogged. Corn plants prefer consistent soil moisture, but standing water can lead to root rot.
 - Weed Control: Remove the weeds visible around the base of the corn plants, as they compete for nutrients and water, potentially adding stress to the corn.

This case highlights the LLM's ability to distinguish between disease and nutrient deficiencies, providing more nuanced analysis than the binary classification alone.

These results demonstrate the complementary strengths of our hybrid approach, combining the rapid, accurate detection capabilities of YOLOv10 with the detailed, context-aware analysis provided by the multi-modal LLM.

DISCUSSION

Our hybrid approach to plant disease detection demonstrates significant advancements in both accuracy and interpretability. The YOLOv10-based model achieved superior performance compared to baseline models, with an mAP50 of 0.87 and an inference time of 15ms per image. This performance can be attributed to YOLOv10's efficiency-accuracy driven design which allows for better feature extraction and utilization, contributing to improved detection accuracy.

The integration of a multi-modal LLM significantly enhances the system's capabilities by providing nuanced, context-aware insights that extend beyond basic disease detection. By offering interpretability and actionable outputs, the system effectively addresses a critical limitation in automated detection systems, as noted by Barbedo (2018).

Our hybrid system has demonstrated several significant implications for agricultural practices:

1. **Early Detection:** The high recall (0.85) enables earlier intervention in disease management
2. **Precision Agriculture:** Location-specific disease information supports precision agriculture practices
3. **Knowledge Dissemination:** The system serves as an educational tool for farmers and agricultural workers
4. **Remote Monitoring:** Efficiency and speed make it suitable for integration with drone or satellite imagery

However, limitations remain. The dataset used may not fully capture the global diversity of crops and regional disease variations, which could impact the system's effectiveness in underrepresented environments. Environmental factors such as lighting and weather, only partially represented in the training data, also pose challenges for real-world deployment. Furthermore, while the LLM enhances interpretability, its computational demands may restrict deployment in resource-constrained settings.

CONCLUSION

This study presents a novel hybrid approach to plant disease detection that combines the strengths of YOLOv10, a state-of-the-art object detection model, with the analytical capabilities of a multi-modal large language model. Our key findings and contributions include:

1. Development of a high-performance YOLOv10-based model for binary classification of plant health, achieving an mAP50 of 0.87 and rapid inference time of 15ms per image.
2. Successful integration of a multi-modal LLM to provide detailed, context-specific analysis of detected diseases, enhancing the interpretability and actionability of results.
3. Demonstration of the system's effectiveness through quantitative evaluation and illustrative case studies, showing its potential for early disease detection and nuanced diagnosis.
4. Analysis of the implications for agricultural practices, including support for precision agriculture and large-scale crop health monitoring.

These advancements represent a significant step towards more effective, accessible, and interpretable plant disease detection systems, with potential impacts on global food security and sustainable agriculture practices.

Future work should focus on addressing the identified limitations and expanding the system's capabilities. Key areas for improvement include:

1. **Multi-Class Detection:** Expanding the classification model to handle multi-class tasks, allowing the detection of specific diseases while retaining strong performance on binary health classification.
2. **Temporal Analysis:** Incorporating temporal data could enable the tracking of disease progression and support predictive capabilities for outbreak management.

3. **Domain Adaptation:** Further improving the system's versatility through domain adaptation techniques, enabling it to generalize effectively to new plant species and geographical regions with minimal retraining.
4. **Model Optimization:** Investigating model compression and optimization techniques will be essential for deploying this system on resource-limited devices, such as mobile phones or field sensors, making it more accessible for on-the-ground use in agriculture.
5. **LLM Enhancement:** Fine-tuning the LLM on domain-specific agricultural knowledge could further enhance the relevance and reliability of its outputs, especially for nuanced diagnoses.
6. **Human-AI Interaction:** Studying human-AI collaboration in agricultural settings would yield insights into how agricultural experts interact with the system, helping refine user interfaces and maximize practical utility in real-world applications.

In conclusion, our hybrid approach to plant disease detection demonstrates the potential for synergistic integration of advanced computer vision techniques and large language models in agricultural applications. As these technologies continue to evolve, we anticipate their increasing role in addressing global challenges in food security, sustainable agriculture, and environmental conservation, making this an exciting and impactful area of ongoing research.

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